RIFAMPICIN

Strong enzyme inducer

Orange-red body fluids

CHEMICAL NAME 3-[(4-methylpiperazin-1-yl)iminomethyl]rifamycin SV (rifampin)
CLASS Rifamycin antibiotic (first-line antituberculosis drug)
TRADE NAMES Rifadin · Rimactane · Rifampin · Rifinah / Rifater (combinations)

THERAPEUTIC USES / INDICATIONS

- Tuberculosis (all forms, combination therapy); latent TB (4 months).
- Leprosy.
- Brucellosis (with doxycycline).
- Staphylococcal bone/prosthetic infections (combination); prophylaxis of meningococcal and *H. influenzae* contacts.

ADMINISTRATION

- **PO** on an empty stomach (30–60 min before breakfast).
- **IV** infusion over 2–3 h when oral not possible.
- Never used alone (rapid resistance) except prophylaxis.

DOSAGES

- **Adult TB:** 10 mg/kg once daily (max 600 mg).
- **Children:** 15 mg/kg (10–20) once daily (max 600 mg).
- **Meningococcal prophylaxis:** 600 mg every 12 h for 2 days (child 10 mg/kg).
- **Brucellosis:** 600–900 mg daily for 6 weeks.

PHARMACOKINETICS

- **Peak:** 2–4 h. Food reduces absorption.
- Metabolised in liver; enterohepatic circulation.
- Excreted mainly in bile. **t½:** 3–4 h (shortens with self-induction).

PHARMACODYNAMICS

- Inhibits bacterial **DNA-dependent RNA polymerase** → bactericidal.
- Potent inducer of CYP3A4, CYP2C9 and P-gp → lowers levels of many drugs.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Orange-red discoloration of urine, sweat, tears (stains contact lenses).
- Nausea, anorexia, abdominal pain.
- Flu-like syndrome (intermittent dosing).

ADVERSE EFFECTS (SERIOUS)

- Hepatotoxicity (jaundice, hepatitis).
- Thrombocytopenia, haemolytic anaemia.
- Acute kidney injury (intermittent dosing).
- Severe skin reactions.

CONTRAINDICATIONS & CAUTIONS

- Rifamycin hypersensitivity; jaundice.
- Use with certain HIV protease inhibitors and other drugs it inactivates (check).
- **Caution:** liver disease, alcohol, porphyria.

KEY INTERACTIONS

- Reduces effectiveness of: oral contraceptives, warfarin, steroids, antiretrovirals, azole antifungals, methadone, phenytoin, sulfonylureas, ciclosporin, tacrolimus, DOACs and many more.
- **Isoniazid, pyrazinamide:** ↑ hepatotoxicity.

NURSING MANAGEMENT

- Baseline LFTs, CBC, renal function; monitor regularly.
- Review every co-prescribed drug for interaction.
- Advise non-hormonal contraception.
- Watch for jaundice, bruising, reduced urine.
- Support daily adherence (DOT).

PATIENT EDUCATION

- Urine, sweat and tears will turn orange-red — harmless, but soft contact lenses may stain.
- The pill/implant will not work; use condoms or IUD.
- Take on an empty stomach every day; do not miss doses.
- Report yellow skin, vomiting or bruising.

KEY POINTS

- RNA polymerase inhibitor — first-line TB (R).
- Orange body fluids.
- Strong enzyme inducer; many interactions (pill fails).
- Hepatotoxic; 10 mg/kg (max 600 mg).

SOURCES CHECKED **[209]** Rifampicin 150 mg Capsules (UK SmPC) **[210]** Rifampin, oral (US label) **[211]** Rifampin, IV (US label)
Full references p. 432–457